CADD Software

A CADD program contains hundreds of functions that enable you to accomplish specific drawing tasks. A task may involve drawing an object, editing an existing drawing, displaying a view of the drawing, printing or saving it, or controlling any other operation of the computer. The functions contain a number of commands that enable you to specify exactly what you want to do and how you want to do it.

The functions are organized into modules that provide easy access to all the commands. Fig.1.5 illustrates a schematic diagram of typical CADD functions in a program. As shown, the program is divided into modules such as draw, edit, data output, function control, data storage and management. A program may also have a number of specialized functions such as layers, database and 3D. Let's have a look at the CADD modules:

- Draw
- Edit
- Data output
- System control
- Data storage and management
- Special features

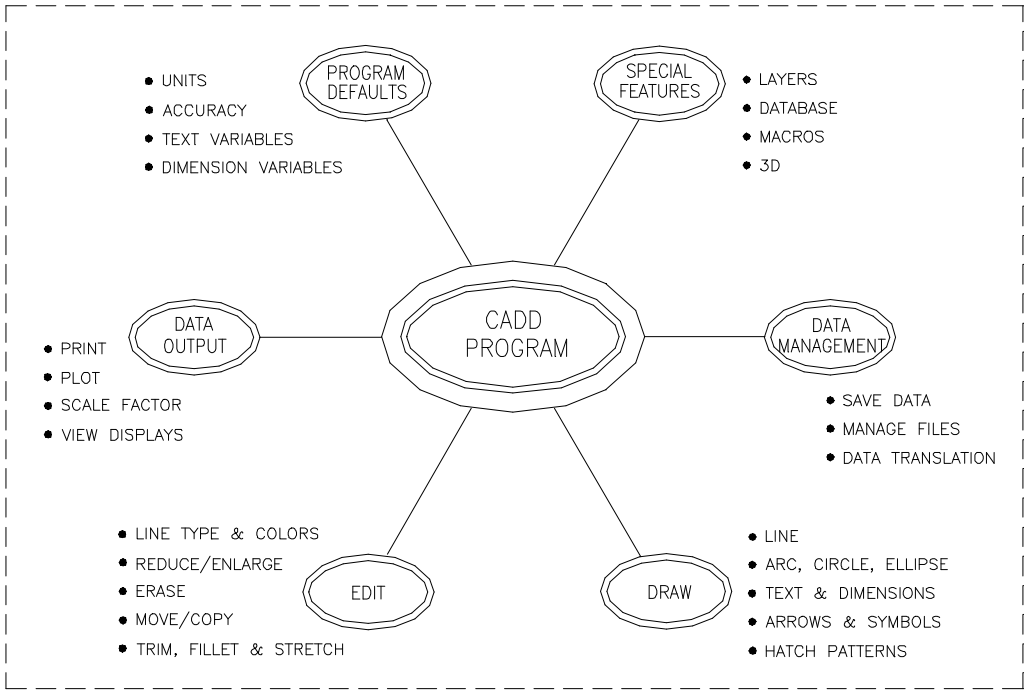

Fig 1.5: *The functions of CADD organized into modules.*

Draw

The draw module provides access to all the drawing functions of CADD. Whenever you need to draw something this group of functions is used. The draw module enables you to draw lines, arcs, circles, ellipses, text, dimensions, symbols, borders and many other drawing components.

Draw is CADD's most frequently used module because all drawing work is accomplished using it.

Edit

The edit module lets you change existing drawing elements and manipulate them in a number of ways. You can move, copy or erase drawing components. You can enlarge or reduce the sizes of diagrams or change the color and line type of drawing components. You can also change the size and style of text and dimensions, as well as edit a dimension to show different units of measurement. A good CADD program is designed to change the appearance of all drawing elements created with CADD.

The edit functions also act as convenient drawing-aid tools. They enable you to join missing corners of lines, trim drawing components along a line, stretch them to fit a new shape, etc. The list of editing capabilities goes on and on. The edit functions make CADD a dynamic drawing tool.

Data Output

The data output module enables you to display drawings on the screen and then print them on paper. There are two separate sets of functions that help accomplish this:

- View-display functions
- Print/plot functions

The view-display functions allow you to display different views of a drawing on the screen. These functions are used quite often, because every time you need to draw something or edit something, you need to focus on that portion of the drawing. With the help of view-display functions, you can zoom in on a specific portion of the drawing.

The print and plot functions allow you to print drawings using a printer or a plotter. You can control many aspects of printing and plotting. You can print the same drawing in different sizes by applying the appropriate scale factor. You can plot the drawings with specific colors, pen thickness, and line types.

Data Storage and Management

The data storage and management module allows you to store and manage drawing data. Through the use of the functions in this module, you can store drawings as files on the hard disk. You can manage the files in directories and sub-directories, and move, copy or delete them as needed.

CADD data management functions also let you translate drawings created by other CADD programs. These functions convert drawing data to a generic format that can be read by any CADD program. Data exchange format (DXF) is one of the common data translation formats used by CADD programs. There are a number of data exchange formats available.

System Control

The system control module (also known as system defaults) allows you to control how CADD works. CADD programs are designed for a broad range of professionals, including architects, designers, engineers and surveyors. With the help of system control functions, you can set the working environment of CADD to suit your needs.

Example: You can set the type of units that you will be using, the accuracy of the units, a style for dimensions and text, colors, layers, line type in a drawing, etc. Additionally, you can customize screen menus, the display of colors on the screen, resolution of the screen, size, the speed of the cursor, etc.

You can also specify whether the selected defaults should apply to a single drawing, to a specific project, or to all the projects in a specific category. The defaults can be set on a temporary or permanent basis.

Special Features

CADD programs usually offer a number of special features that make working with CADD easier and allow you to automate many drawing tasks. For example, you can create layers in a drawing that allow you to segregate drawing components. You can develop spreadsheets and databases that can be used to create many types of project reports. You can create three-dimensional (3D) drawings, such as isometrics and perspectives, with the help of 3D functions. You can also accomplish many other automated tasks with the help of macros.

The number of special features a CADD program has or how elaborate they are varies from one program to another. Some vendors sell specialized features as separate packages, while others include them in a single package. It all depends how a program is written, how big or small it is, and how it is sold.

Important Tip:

The functions of CADD are like the tools of a handyman. All of the tools are essential for working. Some are used more frequently than others are, but each has its own importance.

CADD User Interface

CADD user interface provides the environment and the tools that allow you and the computer to communicate. Each CADD program establishes an environment that best suits its purpose. The goal is to make working with CADD efficient. Most programs use a Graphic User Interface (GUI) to communicate with the user. The GUI provides visual aids for quick data entry. You are given tools to select functions, enter textual or mathematical data, locate points in the drawing window, select objects in the drawing window, etc.

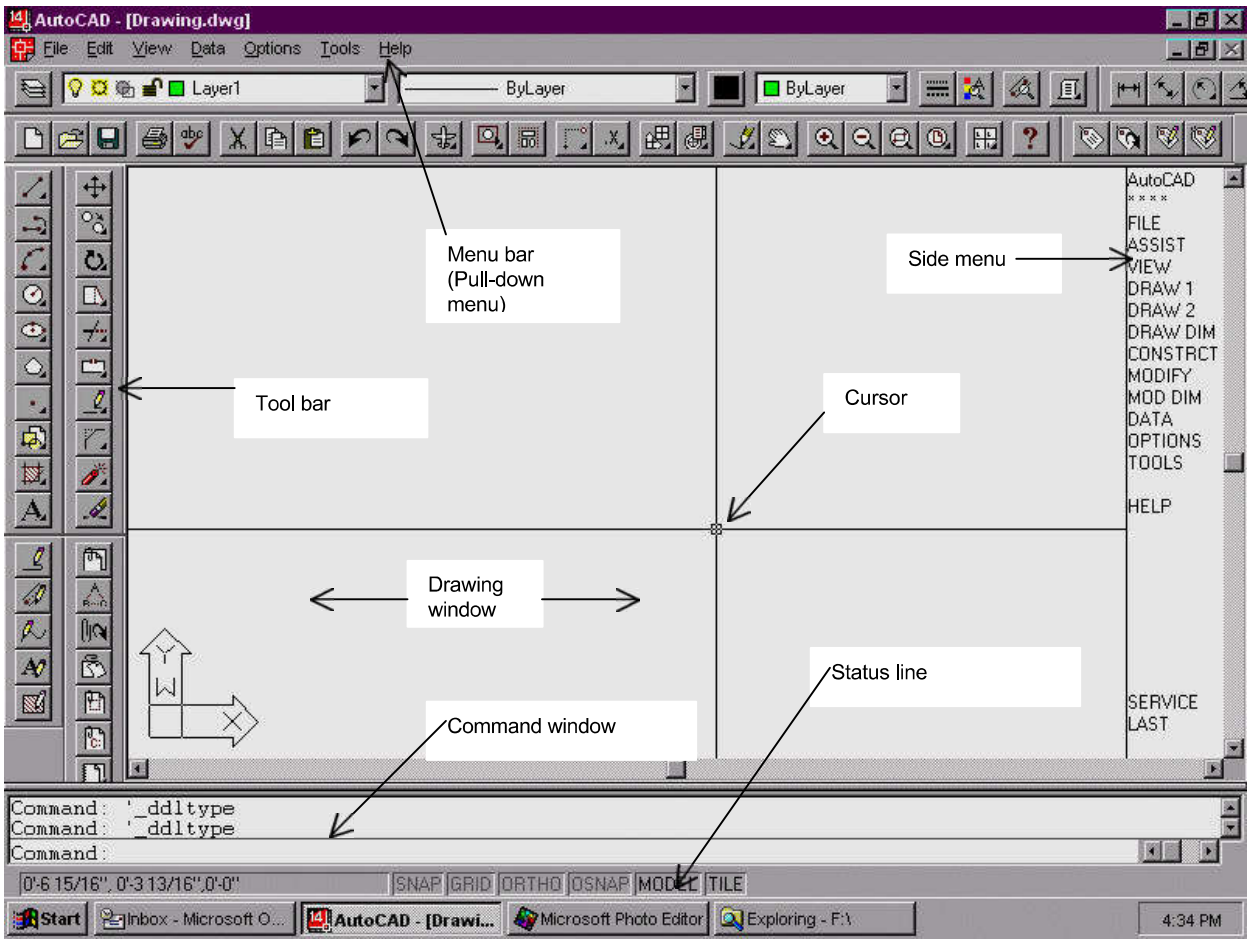

Fig. 1.6: *A graphical user interface used in AutoCAD.*

A typical CADD application window is composed of the following elements: (See Fig. 1.6, an example from AutoCAD)

Command window: An area on the screen that shows all the commands being entered. It also shows what options are available and prompts the user to take action.

Drawing window: The large area in the middle of the application window. All the drawing work is confined within this area.

Menu bar: A horizontal menu located on top that contains the major functions of CADD. When a function is selected from the menu, the menu drops down and displays further options under the menu.

Side menu: A vertical menu that contains all the major functions of the program. To select a function, you need to navigate through the hierarchy of menus and narrow down a selection. The use of side menu has become uncommon in modern CADD programs.

Status line: Shows settings associated with the current drawing on the screen. It shows the current coordinate position of the cursor, current pen number, layer number, etc.

Tool button: A symbol on the screen representing a specific command. When you click on the symbol with the help of the mouse, that particular command is entered.

Tool bar: A collection of tool buttons grouped together. Most CADD programs provide the options to customize the tool bars as required.

The following are the most common ways to communicate with CADD:

- Using the Menu Bar
- Using the tool bars
- Entering commands in the command window
- Working in the drawing area

Using the Menu Bar

Menu bar is an important method of selecting CADD functions. Most CADD programs include this user interface and provide quite similar options. Under the File option, drawing saving and data translation utilities are included. The Edit option allows to cut and paste portions of the drawing and import and export data between various applications. A number of other options in the menu bar allows you to set program defaults and create a customized working environment.

Using the Tool Bars

The tool bars are the latest advancement in the CADD industry. Modern CADD programs use tool bars to provide quick access to commands. The use of tool bars has gained so much popularity that some CADD programs now support only this interface. They don't support command entry through menus, command window or the digitizer.

Using tool bars is a very convenient method of entering commands, because you don't need to type on the keyboard or navigate through the menus. Each command is represented with a specific tool button in the tool bar. To enter a command, all you need to do is click on the tool button with the help of pointing device.

The icons on the tool button suggest the action that the button performs. The exact icons used vary from one program to another, however there are general similarities. Fig.1.7 shows some examples of icons used in CADD. Diagram A shows a drawing tool bar that contains the tool buttons to draw lines, arcs, circle, ellipse, etc. Diagram B shows an editing tool bar that contains the tool buttons for erasing, moving, copying and rotating the drawing elements. Similarly, there are tool bars to display views, printing, plotting drawings and all other CADD functions.

Note:

Most programs allow you to customize the tool bars as needed. You can place frequently used tool buttons in a tool bar, display only specific tool bars, and arrange them on the screen as you like.

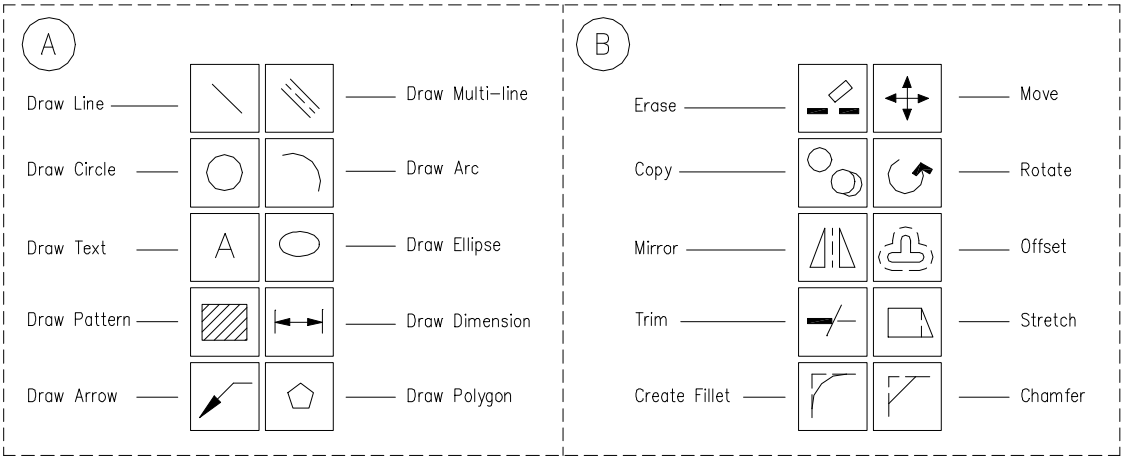

Fig 1.7: *Examples of drawing tool buttons (A) and editing tool buttons (B).*

Entering Commands in the Command Window

The command window is the direct method of entering commands, because you don't need to search through menus. You can enter a command just by typing the command name by using the keyboard. For example, if you want to enter the Line command, simply type the word LINE in the command window and press the Return key and the command is entered.

In many programs, only the initial letter of the command is sufficient to enter the commands. So in our example, only the letter "L" would be required. Most CADD programs allow you to assign any letter or number to a command. You can enter all your favorite commands using these command aliases.

Note:

A drawback of the keyboard entry method is that you need to remember a lot of command names and then type them. An average CADD program has more than 200 commands; it is not always convenient to enter all the commands using the keyboard. Some CADD programs have even eliminated this mode of data entry method.

Working in the Drawing Window

The drawing window is the area where you create a drawing. The drawing window may look small, but it has infinite size like the sky. You can draw as big or as small on this sky like drawing window. The view-display functions allow you to display specific views of the drawing.

When you enter a drawing or editing function, the computer displays the prompt as to where you want that action to take place. You often need to enter points in the drawing window to specify when a drawing element is to be drawn or edited. To locate a point all you need to do is place the cursor at a position where you want to enter the point and then press the Enter button on the puck or mouse.

Important tips:

- It is important to take advantage of all the modes of data entry in CADD and discover which methods work best for you. Most CADD programs offer the same functions available through different interfaces. Some functions you may prefer to access through the command window, others using the menu bar or tool buttons.

- The application window designs are constantly being modified. There are endless possibilities as to how commands can be entered, and how the menus and tool buttons can be used. With some knowledge of programming, you can create your own commands, menus and tool buttons.